

# Text Mining

GIK2KM

**Deadline: May 23, 11:59**

## Description:

The purpose of this lab is to perform text mining operations using natural language processing. You will be performing cleaning, sentiment analysis and classification. You can use this in several applications, few of them are, detecting SPAM, understand positive or negative reviews provided by customers, etc.

## Task1: Sentiment Analysis using polarity

1. You should work with "amazon\_alex.tsv" file.
2. The column of interest for this task is "verified\_reviews". Perform the following cleaning operations to remove:
  - a. symbols, special characters, spaces, numbers
  - b. Tokenize and convert to lower case
  - c. Lemmetize/stem
  - d. Remove stop words
3. Plot the top 20 words based on frequency
4. Now calculate polarity and plot number of negative and positive reviews

## Task2: Text classification using Randomforest

1. You should work with "moviereviews.tsv" file.
2. The column of interest for this task are "label" and "review". Perform the following cleaning operations to remove:
  - a. symbols, special characters, spaces, numbers
  - b. Tokenize and convert to lower case
  - c. Lemmetize/stem
  - d. Remove stop words
3. Plot the top 20 words based on frequency
4. Perform a random split based on 70/30.
5. Use the "Pipeline" function and create a **randomforest classifier**
6. Train the classifier
7. Test and measure the accuracy
8. (Optional) Try to do classification using DistilBERT from Transformers.

\_\_\_\_\_ Hand-in details \_\_\_\_\_

**1.** A video presentation showing the above two tasks should be presented. Maximum time for recording is 10 minutes.

## Things to hand-in

1. Task1: Code and results
2. Task 2: Code and results

### 3. Link to video presentation

**Material:**

Help material for the lab can be found in Canvas – Assignments – Lab 5

**Grade:**

This lab is graded as U/G.

**G** will be given based on below conditions

- Both tasks should be completed successfully along with a video presentation illustrating the tasks.

**Note:** This is a group task.

Good Luck!  
Vijay (vpp@du.se)